



Research Paper

Explainable feature-aware spectral learning for temporally consistent multi-temporal land use/land cover mapping and change detection

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ABSTRACT

Accurate and temporally consistent land use and land cover (LULC) mapping is important for environmental monitoring, ecosystem evaluation, and sustainable land management, especially in mountainous areas with intricate topology and spectral heterogeneity. Nonetheless, most existing machine learning and deep learning approaches have limited spectral interpretability and temporal instability, which undermine multi-temporal change detection. This research study proposes a feature-aware spectral learning framework of pixel-wise multi-temporal LULC classification based on Sentinel-2 surface reflectance data. In a common experimental environment, four pixel-wise spectral learning models, including Feature-Aware Convolutional Neural Network (FA-CNN), Feature-Aware Kolmogorov–Arnold Network (FA-KAN), Feature-Aware Extended Vision Transformer (FA-ExViT), and the proposed Feature-Aware Spectral Vision Transformer (FS-ViT), are comparatively evaluated. While FA-CNN learns local spectral responses, FA-KAN learns nonlinear feature interactions, and FA-ExViT learns global spectral dependencies with self-attention, whereas FS-ViT learns feature-aware spectral tokenization, adaptive spectral reweighting, and coarse-to-fine hierarchical supervision to increase class separability and temporal consistency. The framework is used for multi-temporal LULC mapping 2019, 2021, 2023, and 2025 of the mountainous Coonoor regions. Findings indicate that FS-ViT is consistently superior to comparison models, achieving overall accuracies between 96.46% and 98.07% having Kappa coefficients above 0.93. Explainable artificial intelligence (XAI) exhibits the physically significant and temporally consistent spectral feature contributions. Multi-temporal change analysis shows a net forest loss of 6.31 km², primarily transitioning to low vegetation, alongside vegetation recovery and limited urban expansion. Overall, the proposed FS-ViT framework offers a decipherable and time-resilient solution for long-term LULC mapping and change detection in environmentally sensitive mountainous landscapes.

1. Introduction

Remote sensing imagery is an important data source for environmental monitoring, ecosystem conservation and management, climate impact assessments and land use planning to support sustainable development planning (Ojwang et al., 2024; Waśniewski et al., 2022). Accurate LULC mapping is important for environmentally sensitive areas with complex terrains, as non-homogeneous land-cover patterns, seasonal variations and human impacts exert pressures, making reliable observations with temporally consistent observations (Li et al., 2024). The recent availability of multispectral and multitemporal satellite data has improved the capability to monitor land-surface dynamics at finer spatial and temporal scales (Khan and Sudheer, 2022), but pixel level accuracy and temporal consistency are still problematic. Table 1

describes each LULC category and the associated subcategories used in this work.

Recent advances in deep learning have enhanced LULC classification through a feature learning (Temenos et al., 2023). Convolutional neural networks (CNNs) and encoder–decoder networks perform well in semantic segmentation, but their requirement for spatial patch extraction and fixed receptive fields can reduce pixel-spectral fidelity, limit interpretability, and introduce temporal inconsistencies in multi-temporal analysis (Wang et al., 2022). Transformer based architectures have recently gained attention due to their ability to model long-range dependencies using self-attention (Yao et al., 2023). Even though Vision Transformers demonstrate performance in spectrally heterogeneous environments, the majority of existing approaches act on spatial patches and combine multispectral bands, derived indices, and auxiliary

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Table 1
Description of land use and land cover (LULC) classes and sub-classes used in the study.

Class	Description/Sub-Classes Included
Water Bodies	Lakes, rivers, ponds, reservoirs, wetlands
Urban (Built-up)	Built-up areas, settlements, houses, commercial areas, industrial areas, paved surfaces, roads, transportation networks
Low Vegetation	Grasslands, croplands, agricultural land, shrubs, fallow land, sparse vegetation, plantations
Forest	Dense forests, tree cover, orchards.
Bare Land	Exposed soil, barren land, open fields without vegetation, rocky surfaces, degraded land, and construction sites

variables as one composite input (Khan and Sudheer, 2022). Therefore, explicit modeling of inter-feature spectral relationships and feature-level interpretability modeling.

Temporal semantic consistency is another issue that has been affecting multi-temporal LULC mapping (Li et al., 2024). Many studies generate LULC maps based on independently trained single-date models followed by post-classification change detection of the classification, which can introduce artificial class oscillations. Although explainable artificial intelligence (XAI) methods have been proposed with growing popularity to interpret deep learning models, they are used mostly post hoc, and seldom with a view to measuring feature-level spectral relevance or performance stability over multiple years (Rihan et al., 2025; Temenos et al., 2023). These limitations are pronounced in mountainous regions, where complex terrain and diverse vegetation structures amplify spectral ambiguity and temporal variability (Ojwang et al., 2024). To solve these problems, pixel-wise LULC structures developed to maintain spectral fidelity and explicitly describe feature-aware interactions, offer temporal consistency, and provide interpretable outputs suitable for long-term environmental monitoring.

2. Literature review

Land use and land cover (LULC) mapping has grown to adopt deep learning models to classical machine learning techniques, to enhance spectral heterogeneity and complex landscapes. The initial pixel-wise LULC models were based on the Random Forest (RF) classifier on multispectral bands and auxiliary variables, which showed sufficient accuracy but limited interpretability and temporal resilience (Khan and Sudheer, 2022; Ojwang et al., 2024). Although hierarchical RF-based strategies enhanced the class separability of spectrally ambiguous classes, temporal dynamics have generally analyzed through post-classification comparisons rather than learned representations (Waśniewski et al., 2022). Recent advances increasingly employ deep learning to enhance spectral-spatial discrimination. Modern CNN-based semantic segmentation models showed good pixel-level accuracy, especially on high-resolution images, but the methods that require patches to train and have poor spectral interpretability, which limits their long-term temporal consistency (Wang et al., 2022). Transformer-based models also enhanced the world dependency modeling with models like ExViT, overtly combining multimodal data to perform pixel-

wise LULC classification (Yao et al., 2023). Although effective, the majority of transformer methods work on spatial patches and concentrate on a single date image, but do not explicitly model time and cannot be interpreted at a feature level. Explainable artificial intelligence (XAI) methods are often applied post hoc, rarely embedded within the learning process (Cué La Rosa et al., 2026; Temenos et al., 2023).

Table 2 summarizes LULC classification methods, from traditional machine learning methods to modern deep learning models for different datasets and spatial resolutions. Earlier studies mostly focused on random forest and its hierarchical variants, which demonstrated high classification results on manually defined features, but offered limited interpretability and temporal learning. Recent studies increasingly tend to use CNN-based models, such as U-Net and DeepLab variants, which improve the pixel-level accuracy by capturing spatial patterns. However, these methods often depend on patch-based inputs and lack explicit modeling of spectral dependencies. Some studies have applied XAI techniques, such as SHAP, applied as post hoc analyses but not integrated into the learning process. Moreover, there are still fewer transformer-based and multimodal methods that often consider single-date imagery and do not fully capture the temporal dynamics. These observations suggest that there is a necessity to have a unified framework that can effectively model spectral relationships and ensure interpretability, which is a motivation to take the proposed approach in this study.

2.1. Research gap and contributions

Based on these identified research gaps, the major contributions of this study are presented as follows.

- 1) Existing methods rely on fused feature representations, limiting explicit modeling of inter-feature spectral relationships (Li et al., 2024; Rihan et al., 2025).
- 2) Current LULC frameworks lack mechanisms to ensure temporal consistency across multi-temporal datasets (Farda, 2017; Ojwang et al., 2024; Nguyen et al., 2020).
- 3) Transformer-based and deep learning models generally do not explicitly model inter-feature spectral relationships at the pixel level, reducing their ability to capture meaningful feature interactions (Yao et al., 2023; Wang et al., 2022).
- 4) Patch-based CNN and transformer methods introduce spatial dependency, which compromises pure spectral learning and reduces pixel-level fidelity in heterogenous landscapes (Temenos et al., 2023; Yao et al., 2023).
- 5) Hierarchical learning strategies are underexplored for improving class separability in spectrally complex environments (Waśniewski et al., 2022).
- 6) Explainable AI techniques are mostly applied post hoc and are not used to evaluate the temporal stability of feature importance across multiple years (Rihan et al., 2025; Cué La Rosa et al., 2026; Temenos et al., 2023).

To address the identified research gaps, this study aims to develop a

Table 2
Summary of related LULC studies.

Reference	Dataset/Sensor	Model Used	No. of Classes	Spatial Resolution	Performance (%)
(Khan and Sudheer, 2022)	Landsat TM/OLI	RF	4	30 m	Kappa: 0.93–0.97
(Waśniewski et al., 2022)	Sentinel-2 MSI	Hierarchical RF	9	10 m	OA improvement: +3–9%
(Ojwang et al., 2024)	Landsat-8 OLI	RF (Hierarchical)	35	30 m	OA: 88.1%
(Wang et al., 2022)	Gaofen-2	DeepLabv3+ (CFAMNet)	6	4 m	mIoU: 77.22%
(Li et al., 2024)	Sentinel-2 Time Series	CNN (U-Net) + Indices	8	10 m	OA: 85.01%
(Temenos et al., 2023)	EuroSAT (13 bands)	CNN + SHAP	10	10 m	OA: 87.01%
(Cué La Rosa et al., 2026)	Sentinel-1 & Sentinel-2	CNN (ResNet-18) + XAI	5	10 m	OA: 94.72%
(Rihan et al., 2025)	Landsat (1991–2021)	RF + SHAP	7	30 m	OA: 93.7%
					OA: 88.98–91.53%

feature-aware spectral learning framework for multi-temporal LULC classification. The objective is to design a pixel-wise transformer-based model (FS-ViT) that treats spectral bands, indices, and auxiliary variables as independent tokens to enable explicit inter-feature modeling. The study further aims to incorporate adaptive spectral reweighting to enhance discriminative feature learning and a coarse-to-fine hierarchical supervision strategy to improve class separability and model convergence. In addition, the framework is intended to ensure temporal consistency across multi-temporal datasets through a unified feature representation. This study also included explainable artificial intelligence (XAI) techniques to provide feature-level interpretability and to evaluate the temporal stability of learned representations for reliable LULC mapping and change detection.

3. Study area

Coonoor is a hill station and taluk headquarters located in the Nilgiris District of Tamil Nadu, India, within the ecologically sensitive Western Ghats region (Fig. 1). The study area is located between about 11°20′–11°25′ N latitude and 76°44′–76°50′ E longitude and covers about 223.43 km² after removing cloud-affected and no-data regions. The terrain is mostly mountainous, with an average elevation of ~1850 m above mean sea level and steep slopes, where more than 60% of the area exhibits gradients between 16% and 35%.

Coonoor is a part of a critical montane system that helps in hydro-power generation within the Nilgiris district and is also an essential catchment of a tributary of the Bhavani River. The tea plantations, tourism, and institutional settlements are some of the dominant economic sectors in the region. The climate is subtropical highland, with average temperatures ranging from 13 °C to 27 °C during the summer season (January–April), and annual rainfall of approximately 1670 mm (Saravanan et al., 2021; IMD, 2025), which has resulted in land-use pressure due to rapid urbanization and plantation growth and development, and fragmentation of the habitats, degradation of water quality, and an increase in landslides.

3.1. Data description and preprocessing

Multi-temporal Sentinel-2 Surface Reflectance images from the COPERNICUS/S2_SR_HARMONIZED collection, which used to generate

LULC maps for 2019, 2021, 2023, and 2025. For each year, the images acquired between 1 January and 30 April were selected to maintain Seasonal coherence. This period corresponds to the dry season in the study area, which helps reduce cloud contamination and minimize phenological variability (Nguyen et al., 2020). Multiple images within each time period were combined using a median compositing approach to produce a single representative image for each year. In addition, images with cloud coverage greater than 20% were excluded to improve data quality. All acquired image datasets were uniformly preprocessed to ensure temporal comparability (Waśniewski et al., 2022).

Sentinel-2 multispectral bands were used due to their sensitivity to vegetation, built-up surfaces, water bodies, and bare soil (Fig. 2). SWIR bands were resampled from 20 m to 10 m to match the visible and NIR bands. Surface reflectance was radiometrically normalized, and clouds and cirrus were masked using the QA60 band. All the images were clipped to the study area and standardized to 10 m resolution. Annual cloud-free composites were generated using temporal median reduction to minimize atmospheric effects and sensor noise (Li et al., 2024; Ojwang et al., 2024).

The annual composites were prepared using six spectral bands (B2, B3, B4, B8, B11, and B12). Several spectral indices were generated to improve class separability, namely, NDVI and SAVI to describe vegetation, NDWI to describe surface water and moisture, NDBI, IBI, and BUI to describe built-up and impervious surfaces, and BSI to describe bare soil and sparsely vegetated areas (Temenos et al., 2023). The index formulations are shown in Table 3.

The elevation data of SRTM was used to provide topographical information after being resampled to the Sentinel-2 spatial resolution. The last feature set, a spectral and index-based, and terrain-based feature set, was retained constant across all study years, so that reproducibility of methods was achieved and multi-temporal analysis was possible (Waśniewski et al., 2022).

4. Methodology

4.1. Overall pixel-wise spectral learning framework

This study proposes a hierarchical feature-aware pixel-wise learning framework for land use and land cover (LULC) classification using multi-temporal Sentinel-2 surface reflectance data. The framework is designed

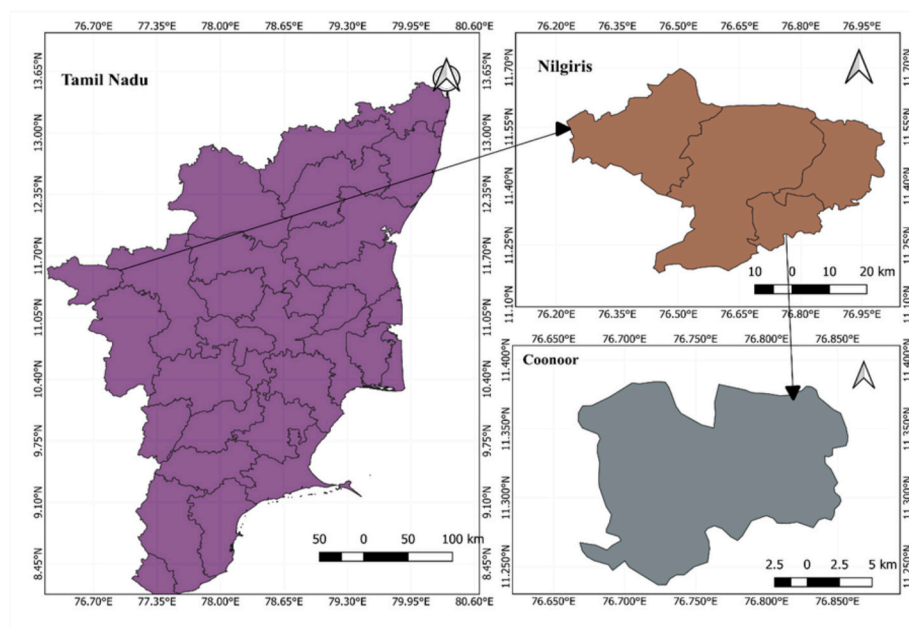


Fig. 1. The location of the study area (Coonoor, Nilgiri).

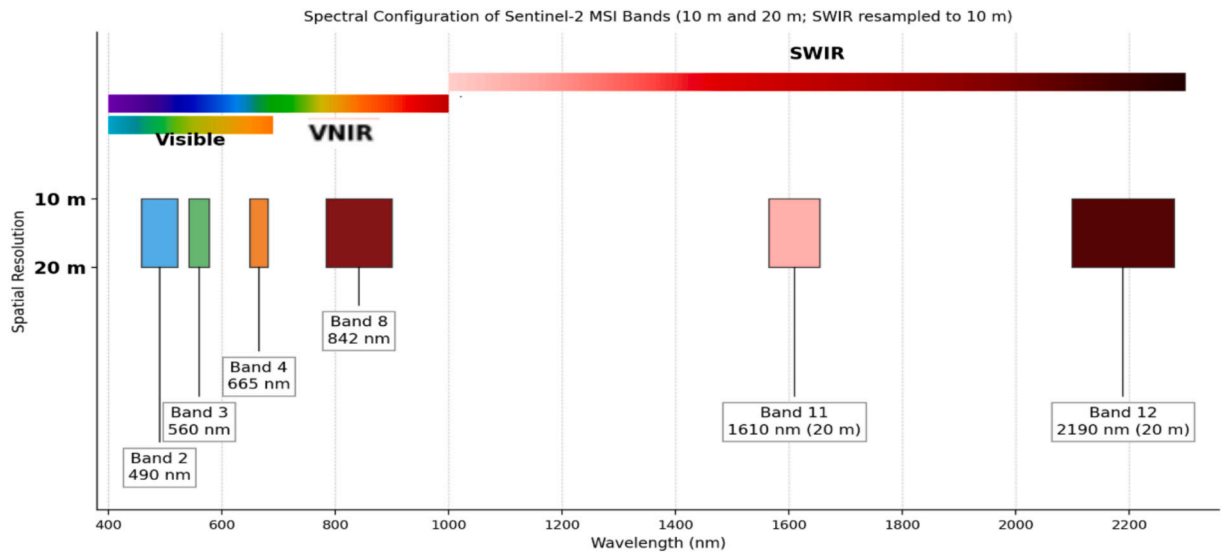


Fig. 2. Spectral configuration of Sentinel-2 MSI VNIR and SWIR bands with corresponding spatial resolutions.

Table 3
Spectral indices used for feature extraction.

S. No	Indices	Formula
1	Normalized Difference Vegetation Index (NDVI)	$NDVI = \frac{NIR - Red}{NIR + Red}$
2	Soil Adjusted Vegetation Index (SAVI)	$SAVI = \frac{(NIR - Red)}{(NIR + RedL)} \times (1 + L)$
3	Normalized Difference Water Index (NDWI)	$NDWI = \frac{Green - NIR}{Green + NIR}$
4	Normalized Difference Built-up Index (NDBI)	$NDBI = \frac{SWIR - NIR}{SWIR + NIR}$
5	Bare Soil Index (BSI)	$BSI = \frac{(SWIR + Red) - (NIR + Blue)}{(SWIR + Red) + (NIR + Blue)}$
6	Index-Based Built-up Index (IBI)	$IBI = \frac{NDBI - (NDVI + NDWI)}{NDBI + (NDVI + NDWI)}$
7	Built-up Index (BUI)	$BUI = NDBI - NDVI$

to be model-agnostic, meaning that it supports multiple learning models under identical experimental conditions, thereby allowing the fair and unbiased comparison among FA-CNN, FA-KAN, FA-ExViT, and the proposed FS-ViT, as shown in Fig. 3. To address the limitations of conventional pixel-wise models in capturing complex inter-feature relationships, FS-ViT is proposed to explicitly model global spectral interactions and learn refined discriminative representations. This enables improved spectral characterization and better class separability within the proposed framework.

Sentinel-2 images that were free of clouds were used to generate multispectral reflectance bands, spectral indices, and auxiliary elevation data. Layers were co-registered and stacked to form a 14-dimensional feature vector at each pixel. Pixels were treated as independent spectral observations, deliberately excluding spatial neighborhood information (Wasniewski et al., 2022), so that performance differences between models reflect solely their capacity for spectral feature modeling rather than spatial context exploitation.

Reference polygons were generated in Google earth engine to extract pixel-wise features and LULC labels. A total of 15,831 labeled samples were exported in a standardized tabular format. The stratified sampling technique was used to divide the dataset into 80% training and 20% validation subsets without any alteration of the distributions of classes (Khan and Sudheer, 2022)

Each pixel p is represented by a feature vector,

$$x_p = [x_{p,1}, x_{p,2}, \dots, x_{p,14}]^T \in \mathbb{R}^{14} \quad (1)$$

comprising six Sentinel-2 reflectance bands (BLUE, GREEN, RED, NIR, SWIR1, SWIR2), seven spectral indices (NDVI, NDWI, SAVI, NDBI, BUI, IBI, BSI), and elevation derived from a co-registered digital elevation model (DEM). Samples with missing or undefined values are excluded.

Feature normalization is performed using standard score normalization,

$$\hat{x} = \frac{x - \mu}{\sigma} \quad (2)$$

where the mean μ and standard deviation σ are derived only from training data and reused during validation and inference in order to prevent information leakage. All the analyzed models have the same input features, training validation split, loss formulation, and optimization settings, ensuring a fair and unbiased comparison.

4.2. Feature-aware spectral tokenization

FS-ViT presents a feature-aware spectral tokenization method that departs from conventional vector-based spectral learning. Rather than combining all features into one embedding, individual spectral bands, index, and auxiliary variables are considered as separate tokens, allowing an explicit model of inter-feature relationships whilst retaining physical interpretability (Yao et al., 2023).

The $F = 14$ scalar features per pixel were linearly mapped to a common embedding space to form a spectral token matrix $Z \in \mathbb{R}^{F \times d}$, where $d = 64$. A classification token that can be learned and positional encodings were added to enable global spectral aggregation and stabilize the training of transformers.

A feature-aware reweighting module was applied before the transformer to enhance discriminative spectral learning. The importance weights of the features were obtained with a lightweight multilayer perceptron on the mean spectral embedding and then the sigmoid activation was performed:

$$w = \sigma(f(\text{Mean}(Z))), Z' = Z \odot w \quad (3)$$

where $f(\cdot)$ represents the learnable MLP. This adaptive reweighting is designed to assign higher importance to informative spectral features while reducing the influence of noisy or redundant responses.

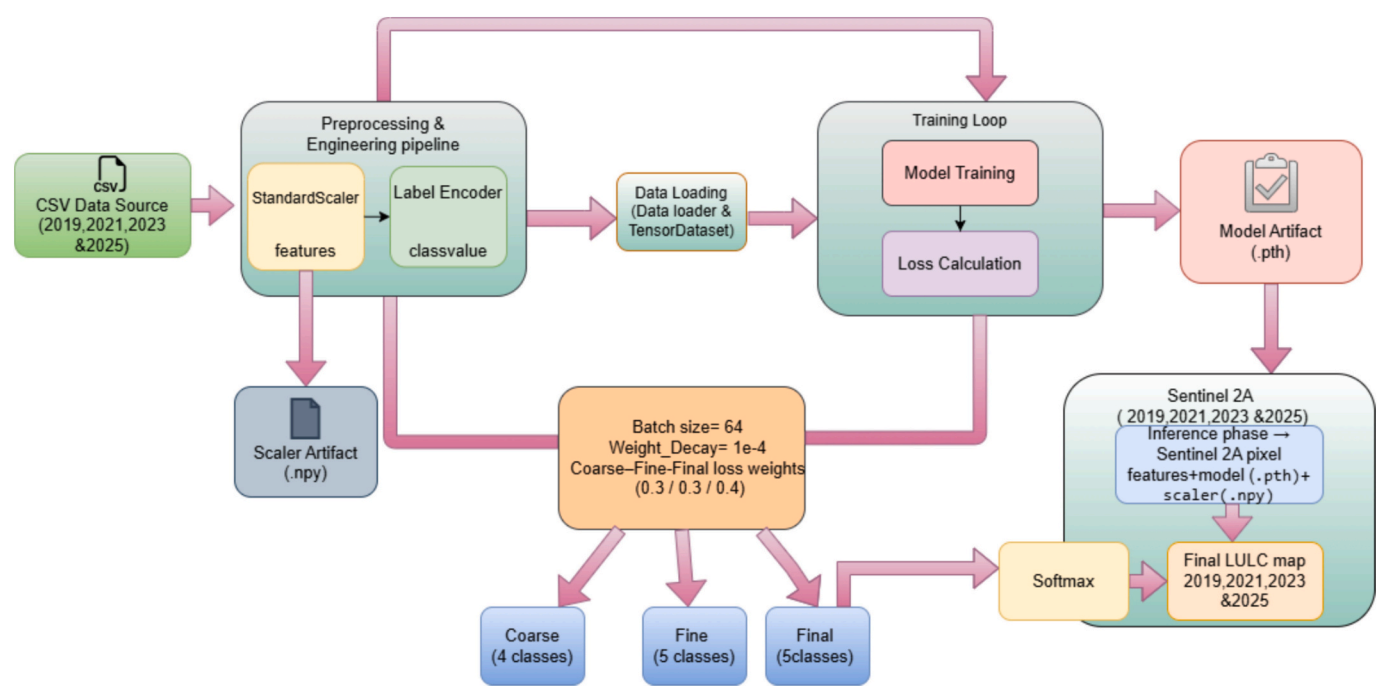


Fig. 3. Overall framework design that shows the idea of remote-sensing input (features based on Sentinel-2) and pixel-wise learning, after which the retrieved features are stored in the form of a table and used to train and make inferences using the model.

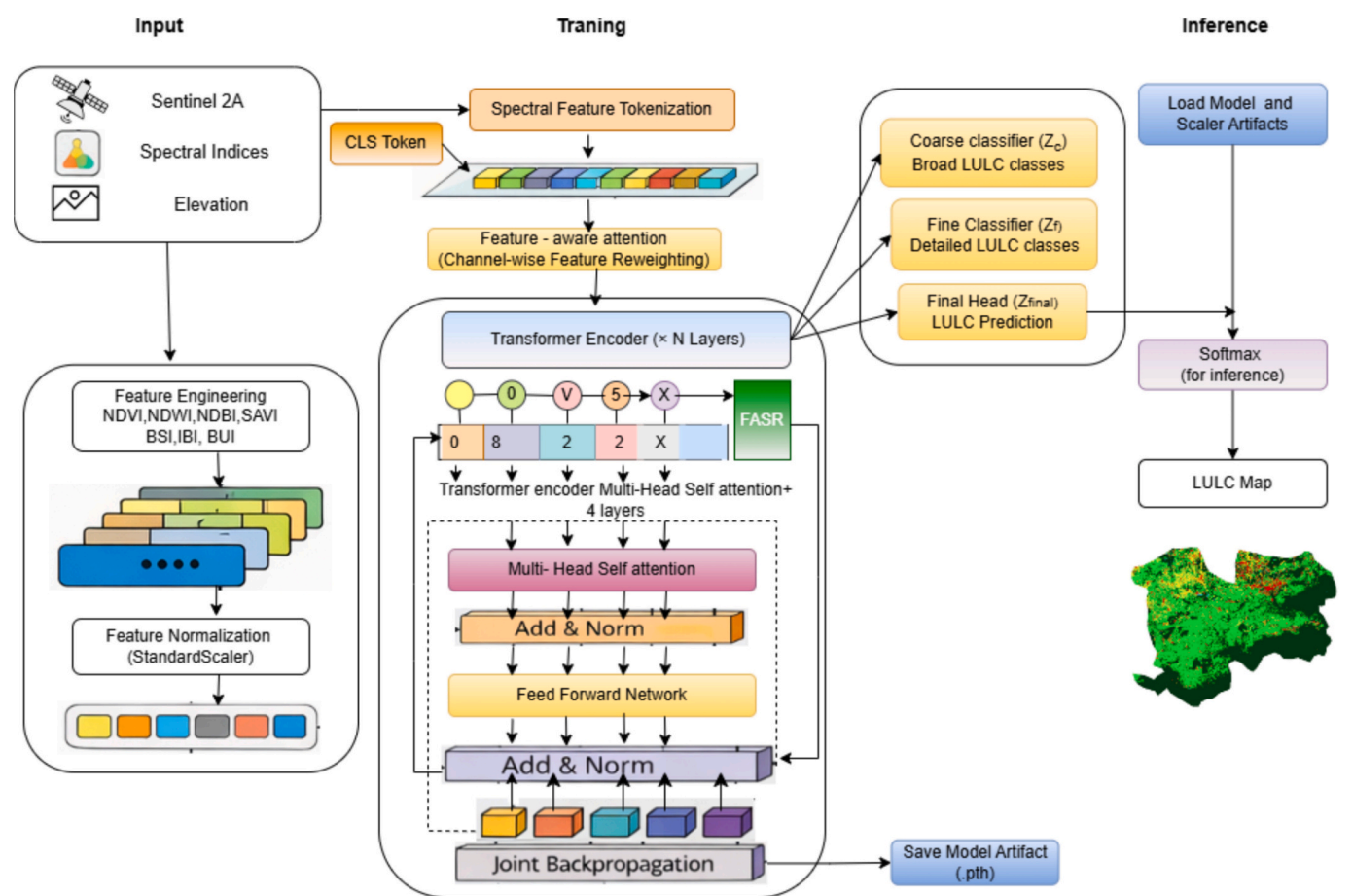


Fig. 4. Feature-Aware spectral vision transformer (FS-ViT) architecture for pixel-wise LULC classification.

4.3. FS-ViT architecture and hierarchical learning

FS-ViT, as illustrated in Fig. 4, uses a transformer architecture that is encoder-only and is used to perform pixel-wise LULC classification with tabular spectral features (Khan and Sudheer, 2022). The re-weighted spectral tokens were fed to a transformer encoder, which had four stacked layers with multi-head self-attention (four heads) and position-wise feed-forward networks with 256 hidden units and GELU. Training stability was done using residual connections and layer normalization.

The self-attention mechanism is defined as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (4)$$

where Q , K , and V represent the query, key, and value projections, respectively.

FS-ViT incorporates a coarse-to-fine hierarchical supervision strategy to guide class representation learning at multiple levels (Ojwang et al., 2024). The spectral similarity was used to put fine-grained LULC classes into four coarse categories, such as water, built-up, vegetation, and bare soil. There were three heads of classification (coarse, fine, and final), which were optimized together with a weighted cross-entropy loss:

$$\mathcal{L} = 0.3\mathcal{L}_{CE}^{\text{coarse}} + 0.3\mathcal{L}_{CE}^{\text{fine}} + 0.4\mathcal{L}_{CE}^{\text{final}} \quad (5)$$

where $\mathcal{L}_{CE}^{\text{coarse}}$, $\mathcal{L}_{CE}^{\text{fine}}$, and $\mathcal{L}_{CE}^{\text{final}}$ are the heads of respective classification. The AdamW optimizer, with a 1×10^{-4} learning rate, batch size of 64, and cosine annealing, was used to train the model in 150 epochs. Only the final head was used during inference.

For each pixel, class logits

$$z = [z_1, z_2, \dots, z_K] \quad (6)$$

were converted to probabilities using softmax, and the final LULC label was assigned via argmax. Pixel-level predictions were reconstructed into georeferenced raster maps in a batch-wise manner (Khan and Sudheer, 2022).

4.4. Models for comparison

To contextualize the performance of the proposed FS-ViT, three feature-sensitive pixel-wise spectral baseline models were implemented and evaluated under the same experimental conditions: FA-CNN, FA-KAN, and FA-ExViT. All models share the same 14-dimensional input feature vector, normalization protocol, train and validation split, optimizer, and learning rate schedule. The key distinction across models lies in the mechanism by which inter-feature spectral relationships are captured.

FA-CNN captures local spectral patterns through channel-wise convolutional weighting, but it has fixed receptive fields and does not model global inter-feature dependencies. FA-KAN approximates nonlinear

feature interactions through learnable activation functions but lacks explicit spectral tokenization, limiting feature-level interpretability. FA-ExViT introduces transformer-based spectral tokenization and self-attention, but applies only generic token-level weighting without spectral-aware reweighting prior to attention computation.

All these limitations are addressed simultaneously in FS-ViT by combining (i) explicit per-feature tokenization, (ii) adaptive spectral reweighting applied before attention, and (iii) coarse-to-fine hierarchical supervision. No single baseline model incorporates all three mechanisms concurrently. Table 4 summarizes the spectral modeling capabilities of all four models.

4.5. Evaluation framework and performance metrics

In order to ensure a fair, consistent, and comprehensive comparison of all models, such as FA-CNN, FA-KAN, FA-ExViT, and the proposed FS-ViT, a single evaluation framework is implemented. Three complementary measures are used to assess classification performance: Overall Accuracy (OA), Kappa coefficient, and spatial map-based accuracy. Together, these metrics quantify global classification correctness, agreement beyond chance, and spatial reliability of the predicted LULC maps.

The multi-metric assessment strategy is especially important for LULC classification tasks, where the presence of class imbalance, spectral ambiguity, and spatial heterogeneity may lead to misleading conclusions if model performance is evaluated using a single metric (Sun et al., 2022).

4.5.1. Overall accuracy (OA)

Overall Accuracy (OA) represents the proportion of precisely classified pixels with respect to the total number of reference samples and is defined as:

$$OA = \frac{\sum_{i=1}^C TP_i}{N} \quad (7)$$

where, TP_i denotes the number of correctly classified samples for class i , C is the total number of LULC classes, and N is the total number of reference samples.

OA provides an intuitive measure of global classification performance; however, it may be biased toward dominant classes in the presence of class imbalance.

4.5.2. Kappa coefficient

The Kappa coefficient (κ) measures the agreement between predicted and reference labels as accounting for agreement that may occur by chance. It is computed as:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (8)$$

Table 4

Spectral modeling capabilities of FS-ViT and baseline comparison models.

Model	Pixel-wise processing	Explicit Spectral tokenization	Feature-aware weighting	Global Spectral dependency modeling	Hierarchical (coarse-fine)	Relative Suitability for Spectrally Complex LULC
FA-CNN	✓	×	✓ (channel-wise)	×	✓	Limited
FA-KAN	✓	×	✓ (Feature-Wise)	✓ (Implicit nonlinear interactions only)	✓	Intermediate
FA-ExViT	✓	✓	✓ (Generic token-level)	✓(self-attention)	✓	Strong
FS-ViT (Proposed)	✓	✓	✓ (Explicit & spectral-aware)	✓ (Coarse-Fine spectral attention)	✓	Strongest

where, p_o represents the observed agreement, and p_e denotes the expected agreement due to random chance.

The Kappa coefficient provides a more conservative and robust evaluation than OA, particularly for multi-class LULC classification problems with uneven class distributions (Sun et al., 2022).

4.5.3. Spatial map-based accuracy assessment

The geographic reliability of the classified land use and land cover (LULC) maps was assessed using a spatial map-based accuracy assessment. An independent set of validation polygons, excluded from model training, was used for unbiased evaluation.

All classified maps were reprojected to a common coordinate reference system and resampled to a spatial resolution of 10 m. Predicted class labels were extracted within the validation polygons using polygon-based spatial sampling.

To obtain a robust estimate of spatial performance, a polygon-level five-fold cross-validation strategy ($K = 5$) was adopted. The validation polygons were randomly divided into five mutually exclusive folds (Farda, 2017). For each fold k , a confusion matrix was generated by comparing predicted and reference labels, from which the Overall Accuracy (OA_k) and the Kappa coefficient (κ_k) were computed.

The final spatial accuracy was reported as the mean and standard deviation across all folds, following standard cross-validation-based performance evaluation approaches (Farda, 2017). The mean Overall Accuracy (OA) and Kappa coefficient (κ) were computed by averaging the fold-wise metrics (OA_k and κ_k) as:

$$OA = \frac{1}{K} \sum_{k=1}^K OA_k, \kappa = \frac{1}{K} \sum_{k=1}^K \kappa_k \quad (9)$$

The variability of the performance across folds was quantified using the standard deviation:

$$\sigma_{OA} = \sqrt{\frac{1}{K} \sum_{k=1}^K (OA_k - OA)^2}, \sigma_{\kappa} = \sqrt{\frac{1}{K} \sum_{k=1}^K (\kappa_k - \kappa)^2} \quad (10)$$

where K denotes the number of folds. These statistical measures provide a robust estimate of classification performance and its stability across different spatial subsets of the validation data.

5. Results and analysis

5.1. Quantitative accuracy comparison

The 2019, 2021, 2023, and 2025 performance of the proposed Feature-Aware Spectral Vision transformer (FS-ViT) was compared with FA-CNN, FA-KAN, and FA-ExViT in terms of Overall Accuracy (OA) and Cohen's kappa coefficient (κ) (Table 5). The training of all models was performed within the same experimental conditions, and thus, the performance difference can only be due to the ability of architectural and spectral modeling (Yao et al., 2023).

As shown in Table 5, FS-ViT consistently achieved the highest classification accuracy across all evaluated years, with peak performance recorded in 2023 ($OA = 98.07\%$; $\kappa = 0.9677$), demonstrating robust spectral discrimination and reliable class separability across varying

land-cover conditions. FA-KAN was in the second place and, therefore, it confirmed the advantages of nonlinear spectral modeling, whereas FA-ExViT showed competitive results, especially in 2023 and 2025, which confirms the efficiency of transformer-based self-attention (Yao et al., 2023); however, its performance remained slightly lower due to the absence of explicit feature-aware spectral weighting. FA-CNN is the lowest accuracy throughout the study period, reflecting the fundamental weakness of convolutions with fixed local receptive fields in capturing global spectral interactions across heterogeneous land-cover scenes (Wang et al., 2022). Although FS-ViT demonstrates competitive classification accuracy compared to FA-ExViT, the proposed model further improves class-wise separability, reduces inter-class confusion, and enhances temporal consistency across years, as evidenced by confusion matrix analysis, spatial LULC maps, and XAI-based feature attribution. These findings suggest that FS-ViT generates more consistent and realistic representations, which are essential for effective multi-temporal LULC analysis.

Fig. 5 shows the model-wise OA trends evaluated across all years. FS-ViT is shown robust, temporally stable and spectral discriminative than a baseline models.

5.2. Class-wise performance and confusion analysis

Confusion matrices at the class level (Fig. 6) indicate that there is a high inter-class confusion in FA-CNN, especially between low vegetation, forest, and bare soil classes. FA-KAN enhances separability by means of nonlinear transformations, and FA-ExViT additionally eliminates confusion with the help of self-attention, but residual errors remain because of the absence of explicit spectral weighting.

Conversely, FS-ViT confusion matrices (Fig. 7) represent high inter-class separability and stable performance by the class, in all years. The analysis of water and urban classes has a near-perfect accuracy, whereas there is a minor amount of confusion between low vegetation and forest that is much lower than in the baseline models. The consistent error trends in a cross-year environment show that FS-ViT is a useful method to address artificial temporal drift.

These classification results are independently confirmed by training and validation loss curves (Fig. 8). These four models for each year effectively converge within 150 epochs, showing that the training and validation losses closely throughout training and further demonstrating that FS-ViT's gains in performance true spectral learning rather than overfitting to training samples.

5.3. Multi-temporal LULC classification maps

Multi-temporal LULC maps generated by all four models for 2019, 2021, 2023, and 2025 are presented in Fig. 9. The top-down arrangement of the maps is as follows: FA-CNN (row 1), FA-KAN (row 2), FA-ExViT (row 3), and FS-ViT (row 4), from left to right in temporal order 2019, 2021, 2023, and 2025.

FA-CNN maps (row 1) show significant salt and pepper noise, including pieces of Bare Soil and Urban patches within the Forest and Low Vegetation zones due to the limited receptive field and inability to model the global spectral context. Urban extents in the FA-CNN outputs change smoothly over years, with interannual differences largely arising

Table 5

Year-wise overall accuracy and kappa coefficient comparison of FA-CNN, FA-KAN, FA-Ex-ViT and FS-ViT Models (2019–2025).

Models	Overall Accuracy				Kappa Coefficient			
	2019	2021	2023	2025	2019	2021	2023	2025
FA-CNN	92.93	93.37	94.06	93.21	0.8777	0.8853	0.8976	0.8829
FA-KAN	95.67	96.05	96.78	96.34	0.9250	0.9319	0.9443	0.9369
FA-Ex-ViT	95.67	95.86	97.03	96.75	0.9250	0.9286	0.9487	0.9439
FS-ViT (Proposed)	96.46	97.28	98.07	97.66	0.9388	0.9532	0.9677	0.9597

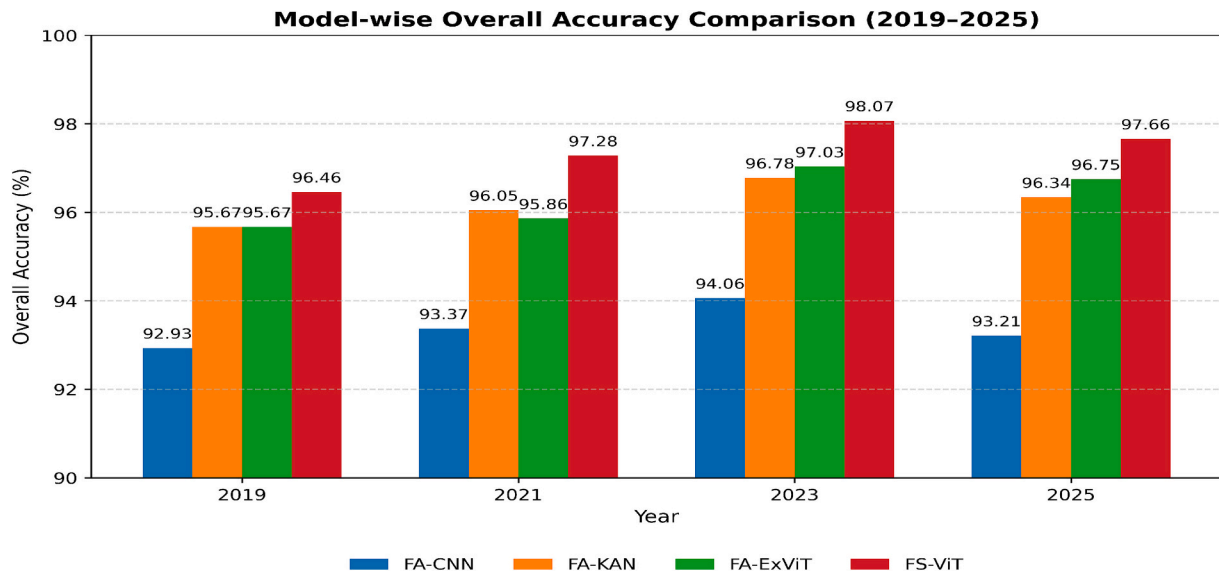


Fig. 5. Model-wise comparison of Overall Accuracy (%) for the FA-CNN, FA-KAN, FA-ExViT, and proposed FS-ViT models across the years 2019, 2021, 2023, and 2025.

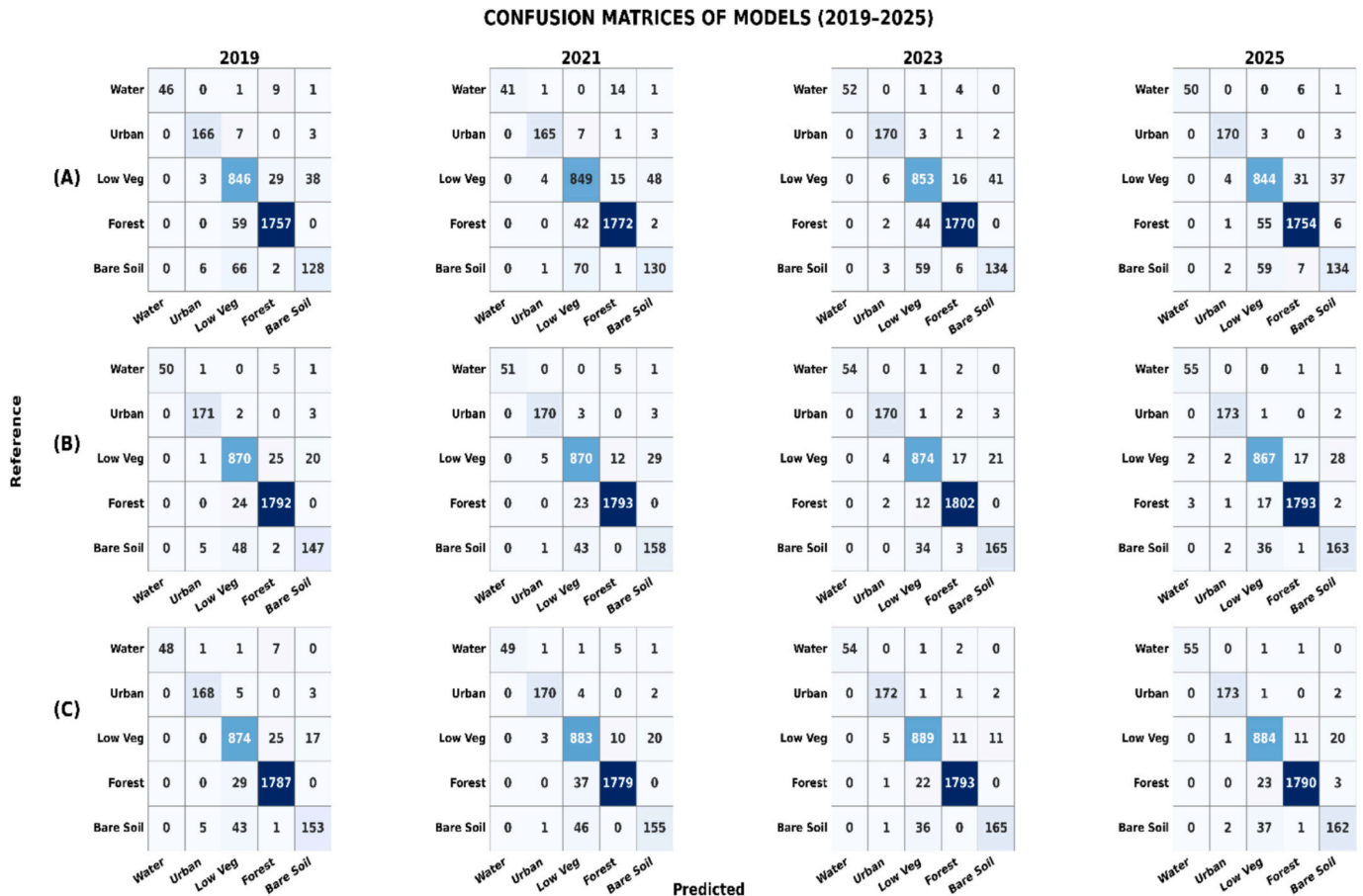


Fig. 6. Comparative confusion matrices of FA-CNN (A), FA-KAN (B) and FA-ExViT (C) for 2019, 2021, 2023, and 2025. Rows arrange model architectures of FA-CNN, FA-KAN and FA-ExViT from top to bottom; columns show temporal snapshots left to right (2019, 2021, 2023, 2025), making direct cross-model and cross-temporal comparison in class-wise classification performance.

from misclassification rather than real land-cover change. FA-KAN maps (row 2) are spatially smoother. Urban class boundaries are still imprecise at transition zones, with visible over-prediction of urban extent, particularly in 2021 and 2023. FA-ExViT maps (row 3) show further

improvement in spatial coherence; however, subtle Urban class over-representation is still discernible in the northern sector of the study area across multiple years.

FS-ViT maps (row 4) exhibit the highest spatial coherence across all

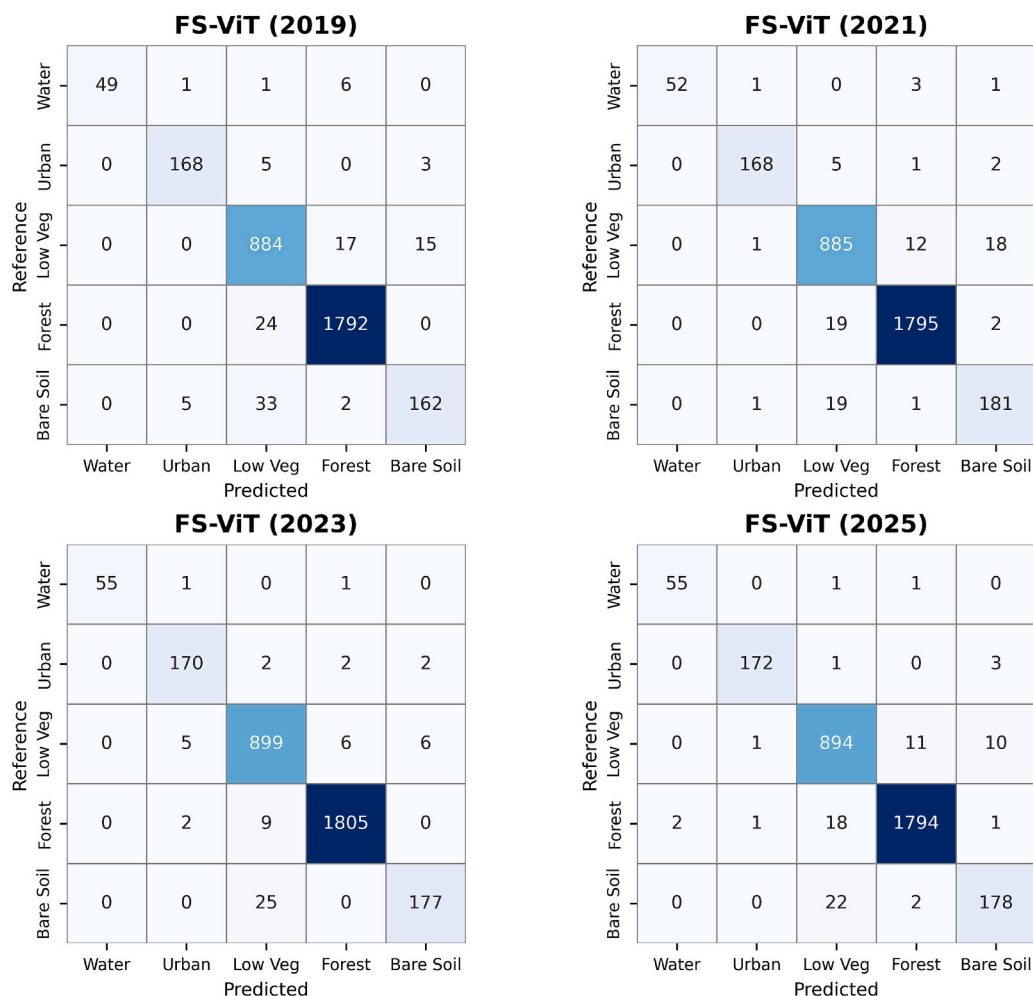


Fig. 7. Confusion matrices of the proposed FS-ViT model for the years 2019, 2021, 2023, and 2025, showing strong diagonal dominance and consistent class-wise performance over time.

years, with well-defined class boundaries, minimal noise, and consistent class distributions. The Urban class in FS-ViT outputs is consistent and well defined than in any baseline model, a direct consequence of the feature-aware spectral weighting mechanism that distinguishes built-up impervious surfaces from spectrally overlapping Bare Soil and Low Vegetation. As a result, the difference between models with respect to Urban extent perceptible for rows of Fig. 9 is a true reflection of architectural performance differences that is fully consistent with the quantitative accuracy result outlined in Table 5 and the confusion matrix analyses for Figs. 6–7.

5.4. Spatial accuracy and temporal consistency assessment

A polygon-level five-fold cross-validation methods ($K = 5$) were used to assess the spatial accuracy of classified LULC maps. An independent set of validation polygons were used for unbiased evaluation that did not overlap any polygons used during model training or testing. These validation polygons were randomly divided into five mutually exclusive and non-overlapping folds to reduce spatial autocorrelation effects. Each year, there were about 4107 validation pixel samples, with an average of 821 samples per fold.

Table 6 summarizes the spatial accuracy of FS-ViT. In 2019, the model performed with a mean OA of 0.9630 and a k of 0.9318, indicating that it agrees well geographically between predicted and reference labels. The highest performance was achieved in 2021 (mean OA = 0.9707; $k = 0.9496$) and 2023, when the mean OA was 0.9765, and the

standard deviation was the lowest ($SD = 0.0096$), indicating a high level of consistency in predictions between folds. At spatial performance, in 2025, the mean OA is 0.9727, and k is 0.9543. High consistency between folds and low dispersion indicate that FS-ViT delivers geographically consistent and temporally stable maps of LULC, which shows that the model is suitable to be used in long-term measurements in mountainous areas.

5.5. LULC change analysis

Land-cover transition in the proposed FS-ViT model is coherent, and the temporal dynamics of class-wise land use and land cover between the years 2019 and 2025 at a given stable spatial extent of 223.44 km². The observed changes can be interpreted as real land-cover changes and not geometrical or preprocessing artifacts because a constant mapped area is being used, therefore. Multi-temporal LULC maps (Fig. 10) generated using the proposed FS-ViT model demonstrate high temporal consistency, low inter-year class oscillation, and stable spatial distribution patterns across the study period (2019–2025).

Table 7 summarizes the quantitative FS-ViT class-wise area statistics of Table 6. The largest net change in forest cover is a negative change of 104.52 to 98.22 km² from 2019 to 2025 (−6.31 km²). Low vegetation, on the other hand, gains (+3.61 km²), suggesting the redistribution of vegetated surfaces and partial recovery of vegetated surfaces. Urban regions have a low growth rate in the area (+0.44 km²), and bare land grows by (+2.15 km²), with a significant interannual range. Water

Training vs Validation Loss (2019–2025)

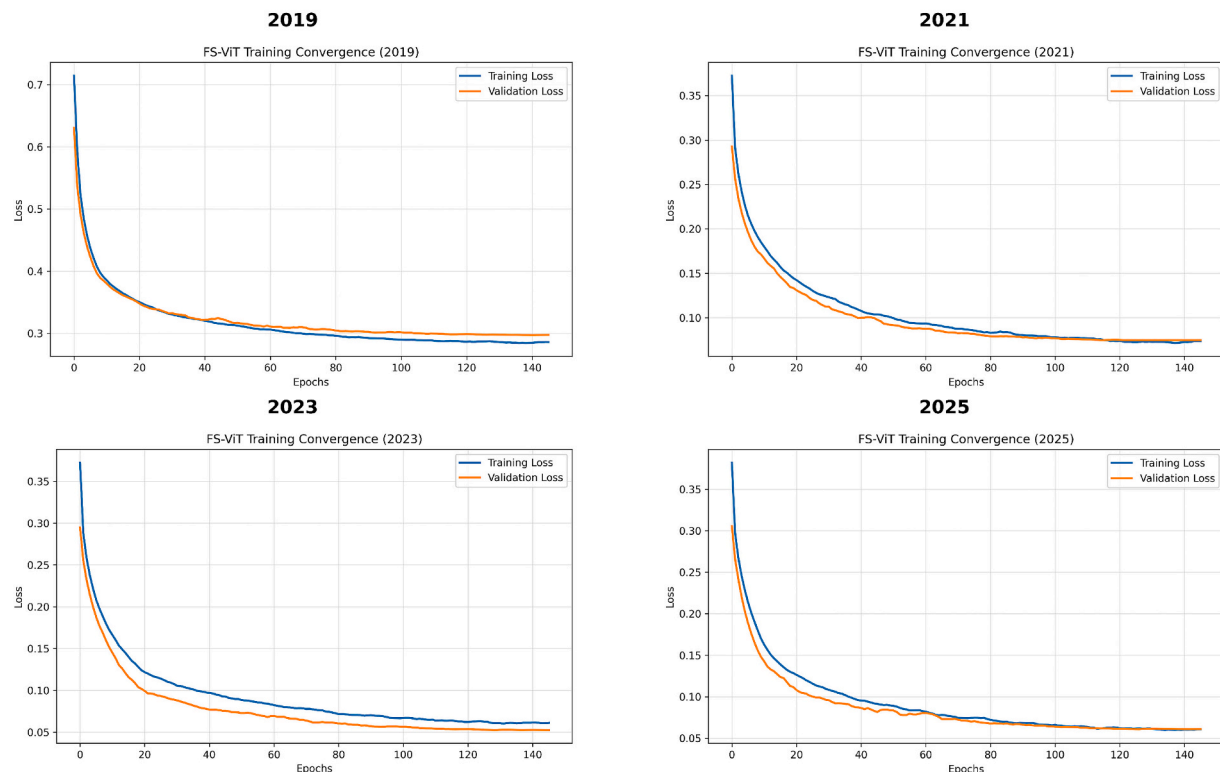


Fig. 8. The proposed FS-ViT model's training and validation loss curves for the years 2019–2025 show steady convergence and reliable generalization.

bodies change slightly over the study period ($+0.11 \text{ km}^2$). These trends are characterized by a relatively slow anthropogenic environmental change, including forest degradation, redistribution of vegetation, and gradual urban expansion that influences the stability of ecosystems in the Western Ghats (Khan and Sudheer, 2022; Yao et al., 2023).

In order to analyze these dynamics in more detail, a pixel-based LULC transition analysis between 2019 and 2025 was performed (Table 8). The most important change was the conversion of forest to low vegetation (14.77 km^2), suggesting progressive degradation rather than immediate deforestation. The concurrent changes, from low vegetation to forest (9.66 km^2) and from bare soil to low vegetation (4.08 km^2), illustrate that re-vegetation continues, and land recovery is still going on. This dynamic includes conversion of low vegetation (5.24 km^2) to bare soil and forest (0.99 km^2) to bare soil, which suggest short term land clearing, urban expansion is spatially constrained mainly resulting from low vegetation (1.69 km^2) change with a minimal effect of direct forest to urban (0.87 km^2).

5.6. Spatial patterns of LULC transitions

The quantitative transition pathways shown in Table 8 are further validated using representative spatial patterns derived from FS-ViT classification maps. These transition dynamics across heterogeneous landscape zones in Fig. 11A and B, where classified maps are presented next to high-resolution imagery for direct visual validation of classification outputs.

Forest and low vegetation areas in degradational zones (Fig. 11A) show a progressive transition driven by varying levels of disturbance. Forest areas transition directly to bare soil (Row A), indicative of significant disturbance processes, including selective logging and terrain modification. Forest areas also fragment into low vegetation (Row B), indicating intermediate degradation following canopy thinning, without

complete vegetation loss. Also, low vegetation patches bare soil transition further into bare soil (Row C), indicating surface degradation and active land clearing. These three transition pathways demonstrate a gradient of degradation severity ranging from partial canopy loss to full vegetation removal.

A reverse spatial pattern is observed in recovery zones (Fig. 11B, Rows E and F). Bare soil patches transition to low vegetation cover (Row E) in 2025, and areas with low vegetation are leading to the regeneration of forest (Row F), which also reflects active natural recovery processes. Quantitative evidence for these transitions is provided by the conversion of 9.66 km^2 from low vegetation to forest and 4.08 km^2 from bare soil to low vegetation (Table 8).

The results in urban encroachment zones (Fig. 11B, Row D) show that built-up areas have expanded incrementally, primarily from bare soil regions (as illustrated in Row D), and also from low vegetation areas, with minimal direct intrusion into adjacent forest patches. This observation is consistent with the limited forest-to-urban transition (0.87 km^2) shown in Table 8, indicating that urban growth is largely confined to already modified landscapes.

Overall, the strong correspondence between spatial patterns is shown in Fig. 11 and the quantitative transitions observed in Table 8, as well as class-wise trends shown in Table 7, confirms that FS-ViT is capable of capturing both degradation dynamics and recovery processes with a high spatial fidelity and temporal consistency over highly heterogeneous mountain environments.

5.7. Explainable artificial intelligence (XAI) analysis

To improve interpretability, post-hoc XAI analysis was performed to assess each spectral feature contribution to LULC classification in FS-ViT. A gradient-based attribution method ensures the feature importance without altering the model. Latent representations were also

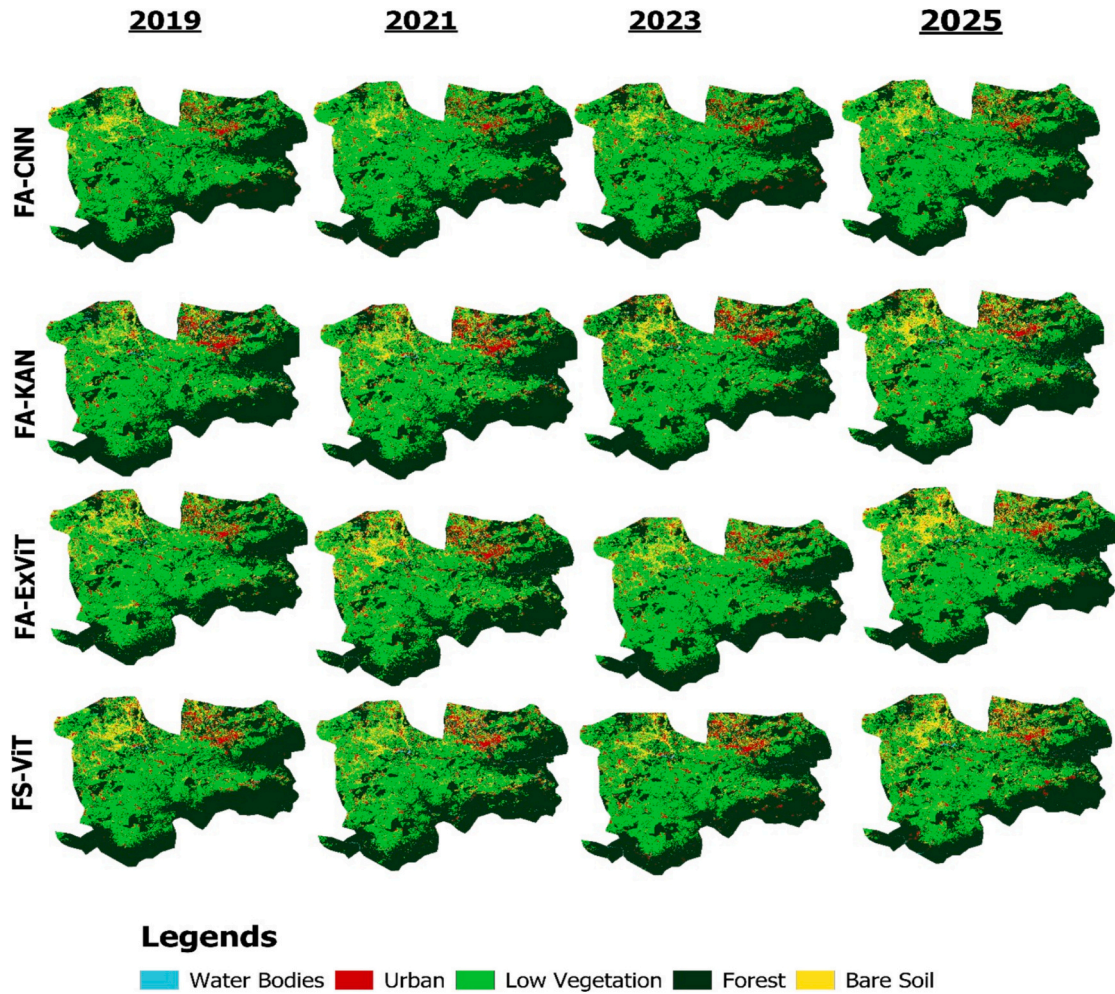


Fig. 9. Multi-temporal LULC maps (2019–2025) generated by FA-CNN (row 1), FA-KAN (row 2), FA-ExViT (row 3), and the proposed FS-ViT (row 4). Maps are arranged in top-to-bottom model order and left-to-right temporal order (2019, 2021, 2023, 2025).

Table 6
Spatial accuracy assessment of FS-ViT (2019–2025).

Year	Mean OA (K-fold)	Std OA	Mean Kappa	Std Kappa
2019	0.9630	0.0242	0.9318	0.0386
2021	0.9707	0.0181	0.9496	0.0292
2023	0.9765	0.0096	0.9568	0.0175
2025	0.9727	0.0134	0.9543	0.0235

analyzed to assess class separability, revealing spectral sensitivity, discriminative behavior, and temporal robustness.

5.7.1. Class-wise feature attribution analysis

An analysis of the decision-making behavior of the proposed FS-ViT model was performed through a post-hoc, gradient-based explainable artificial intelligence (XAI) strategy to measure the value of each spectral band and derived indices in making land use and land cover (LULC) predictions, without changing the parameters of the trained model (Rihan et al., 2025). The importance of features was calculated in Gradient x Input, which is applied to the final classification head of FS-ViT. Given an input sample x and the predicted class c , the attribution score of each feature was:

$$\text{Importance}(x_i) = |x_i \cdot \frac{\partial y_c}{\partial x_i}| \quad (11)$$

where, y_c represents the logit value of class c that is predicted. Where y_c represents that value of y_c in representative samples of each class, and therefore y_c is normalized to allow reasonable comparison between spectral features. To evaluate temporal consistency in feature relevance, class-wise attribution analysis was performed separately for the years 2019 and 2025 (Fig. 12), allowing direct comparison of spectral importance patterns across time.

- **Water bodies** are primarily influenced by *NDWI*, *BLUE*, and *GREEN*, reflecting strong absorption and reflectance properties of water surfaces.
- **Built-up areas** show notable contributions from *SAVI*, *NDVI*, and elevation, along with moderate influence from *SWIR* bands, indicating spectral overlap between impervious surfaces and surrounding vegetation.
- **Low vegetation** is mostly characterized by *NDVI*, *SAVI*, and *RED*, signifying average vegetation cover and partial soil exposure.
- **Forest regions** are highly dependent on *NIR*, *NDVI*, and *SAVI*, reflecting dense canopy cover and advanced photosynthetic activity.
- **Bare soil** shows higher sensitivity to *BSI*, *RED*, and *SWIR*, corresponding to variations in soil brightness and moisture content.

In both years, the LULC classes exhibit a high level of stability over time in the relevance of features, and both show only slight differences due to seasonal or environmental factors. This uniformity signifies that FS-ViT framework models physically significant and temporally stable

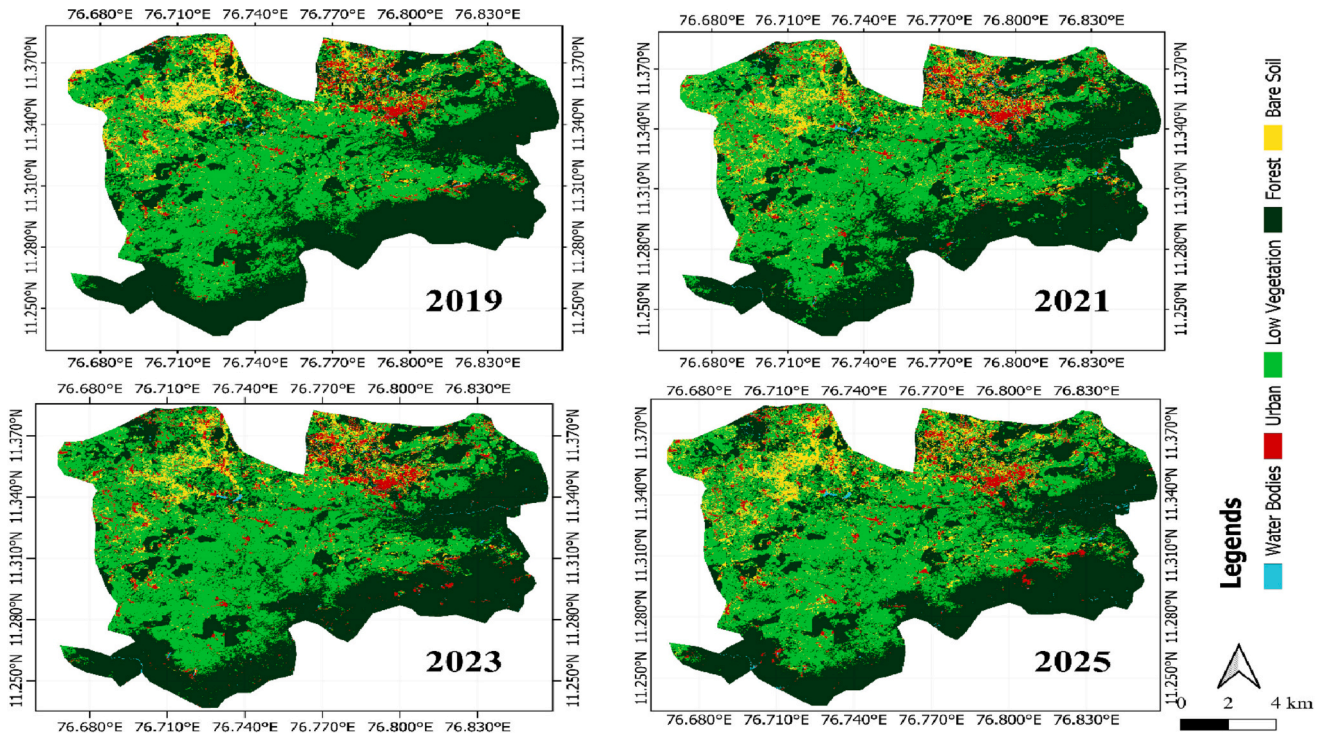


Fig. 10. Multi-temporal land use and land cover (LULC) maps (2019–2025) were generated using the proposed FS-ViT models.

Table 7

Class-wise land use and land cover (LULC) area statistics (km²) derived from FS-ViT classification for the years 2019, 2021, 2023, and 2025, along with net changes over the study period (2019–2025).

Class	2019	2021	2023	2025	$\Delta(2019-2025)$
Water	0.44	1.12	0.47	0.55	+0.11
Urban	9.84	10.73	11.12	10.28	+0.44
Low Veg	96.17	95.59	98.21	99.78	+3.61
Forest	104.52	103.34	103.95	98.22	-6.31
Bare Soil	12.46	12.66	9.68	14.61	+2.15

Table 8

Pixel-wise land use and land cover (LULC) transition matrix (2019–2025) derived from FS-ViT. Values represent the area converted between classes (km²).

2019\2025	Water	Urban	Low Veg	Forest	Bare Soil	Total Loss
Water	0.00	0.03	0.07	0.12	0.00	0.22
Urban	0.00	0.00	1.37	0.00	1.17	2.54
Low Veg	0.00	1.69	0.00	9.66	5.24	16.59
Forest	0.00	0.87	14.77	0.00	0.99	16.63
Bare Soil	0.00	0.66	4.08	0.86	0.00	5.60
Total Gain	0.11	3.22	20.29	10.52	7.40	–

spectral attributes, other than depending on chance associations (Temenos et al., 2023). These findings further validate the effectiveness of feature-aware spectral learning in capturing stable and interpretable representations for multi-temporal LULC analysis.

5.7.2. Latent feature space analysis using UMAP

The latent embeddings of the last transformer encoder layer were extracted and Uniform Manifold Approximation and Projection (UMAP) were used to further study the internal representations learned by FS-ViT. The embeddings are related to the CLS token, which expands high-level spectral-semantic information that is acquired during training. UMAP was chosen because it is able to maintain the local neighborhood relationship and the global data organization, thus

making it appropriate for visualizing complex feature manifolds.

Two levels of class abstraction were analyzed:

- **Coarse-level classes:** Water, Urban, Vegetation, Bare soil
- **Fine-level classes:** Water, Urban, Low vegetation, Forest, Bare soil

The coarse “Vegetation” class aggregates categories, while fine-level classification separates it into low vegetation and forest. This hierarchical strategy enables evaluation of both global separability and fine-grained discrimination. Using identical classes would yield similar clusters but fail to capture intra-class variability within vegetation.

UMAP projections at coarse-level (Fig. 13a) for the year of 2019 show well-separated, compact clusters for all land-cover classes. Distinct spectral properties cause it to be very separable from water, urban and bare soil are also well distinguished. Vegetation is a single, dense group with high intra-class consistency. At the fine, vegetation (Fig. 13a) separates into low vegetation and forest. The distribution remains structured, and some overlap exists, but it is reflecting ecological similarity and transition zones. This confirms that FS-ViT preserves meaningful relationships among spectrally similar classes while maintaining overall class separability.

UMAP projections for 2025 (Fig. 13b) show stable vegetation and water clusters, while urban and bare soil shows higher dispersion due to land changes. Fine level separation between forest and low vegetation improves. The FS-ViT maintains a globally stable but locally adaptive feature (Fig. 13 a-b) from 2019 and 2025, represents the temporal robustness and environmental sensitivity. These results exhibit that FS-ViT not only achieves high accuracy in classification but also learns features with structural meaning and semantic coherence. The UMAP analysis provides supporting evidence that our proposed feature-aware spectral learning framework can learn representations that effectively capture both global class separability and fine-grained intra-class variability, supporting its robustness for multi-temporal LULC classification.

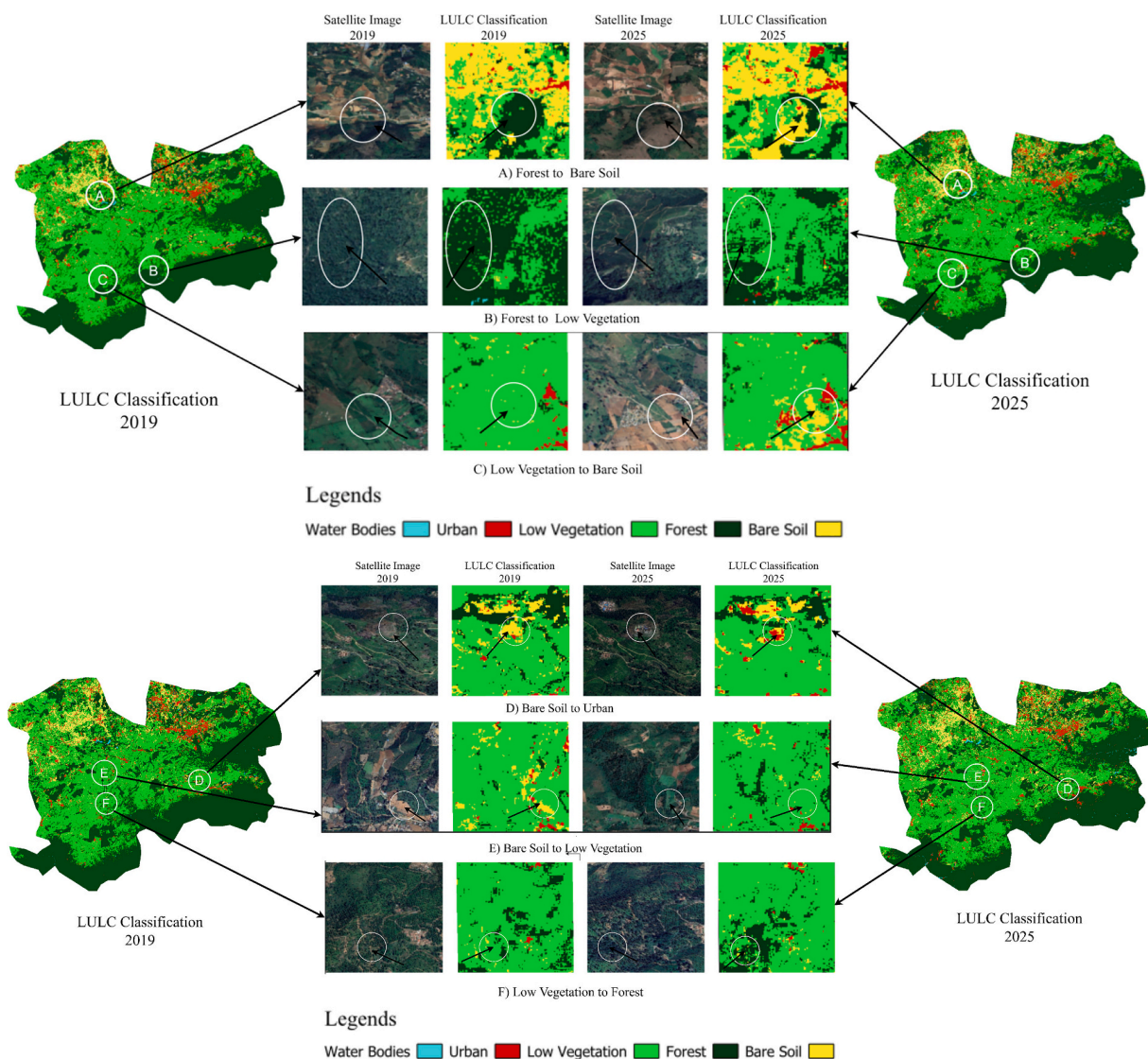


Fig. 11. A. Representative spatial patterns of dominant LULC transition pathways (2019–2025) derived from FS-ViT classification. Rows illustrate (A) forest to bare soil, (B) forest degradation to low vegetation, and (C) low vegetation to bare soil across heterogeneous landscapes. The left and right overview maps show the full study area classifications for 2019 and 2025 respectively. Each row pairs high-resolution satellite imagery with corresponding classified maps for 2019 and 2025; annotated regions of interest are indicated by white circles with arrows. B. Continued representative spatial patterns of dominant LULC transition pathways (2019–2025), illustrating (D) bare soil to urban expansion, (E) bare soil to low vegetation, and (F) low vegetation to forest regeneration in ecologically transitional zones. The left and right overview maps show the full study area classifications for 2019 and 2025, respectively. Each row pairs high-resolution satellite imagery with corresponding classified maps for 2019 and 2025; annotated regions of interest are indicated by white circles with arrows.

6. Discussion

The proposed FS-ViT framework outperformed FA-CNN, FA-KAN, and FA-ExViT in all of the study years with overall accuracies ranging from 96.46% to 98.07%, Kappa coefficient values >0.93 (Table 5). This was also confirmed with polygon-level five-fold cross-validation, achieving a mean OA equal to 0.9765 and a relatively small standard deviation of accuracy values equal to 0.0096 in 2023 (Table 6). These results further confirm that explicit feature-wise spectral tokenization, adaptive reweighting, and coarse-to-fine hierarchical supervision together improve class separability and temporal stability—especially for spectrally similar classes such as low vegetation, forest, and bare soil (Yao et al., 2023; Waśniewski et al., 2022; Rihan et al., 2025).

The multi-temporal change analysis indicated a net forest decline of 6.31 km^2 between 2019 and 2025 (Table 7), with the most common transition being from forest to low vegetation (14.77 km^2), while recovery signals were dominant moving from low vegetation to forest

(9.66 km^2) and bare soil to low vegetation (4.08 km^2) (Table 8), indicating simultaneous degradation and regeneration processes coexisting in this area. Urban growth continued to be spatially limited ($+0.44 \text{ km}^2$) during the study period (Table 7), mostly contained to already modified landscapes, in accordance with the low-impact anthropogenic transition of the Western Ghats (Khan and Sudheer, 2022; Yao et al., 2023).

Based on XAI-based gradient attribution, represent that the FS-ViT captured spectrally stable temporal patterns in both years that were meaningfully defined across biophysical terrain types (Fig. 12), and the latent embeddings analysis through UMAP confirmed well-separated class clusters and temporally stable feature organization (Fig. 13a–b), confirming the model as an interpretable machine learning technique responding consistently over time at multiple scales (Temenos et al., 2023; Cué La Rosa et al., 2026). However, the framework does not explicitly incorporate spatial context and involves a higher computational cost. Future work will focus on integrating spatial-spectral fusion, multi-source data, and spatio-temporal transformer architectures to

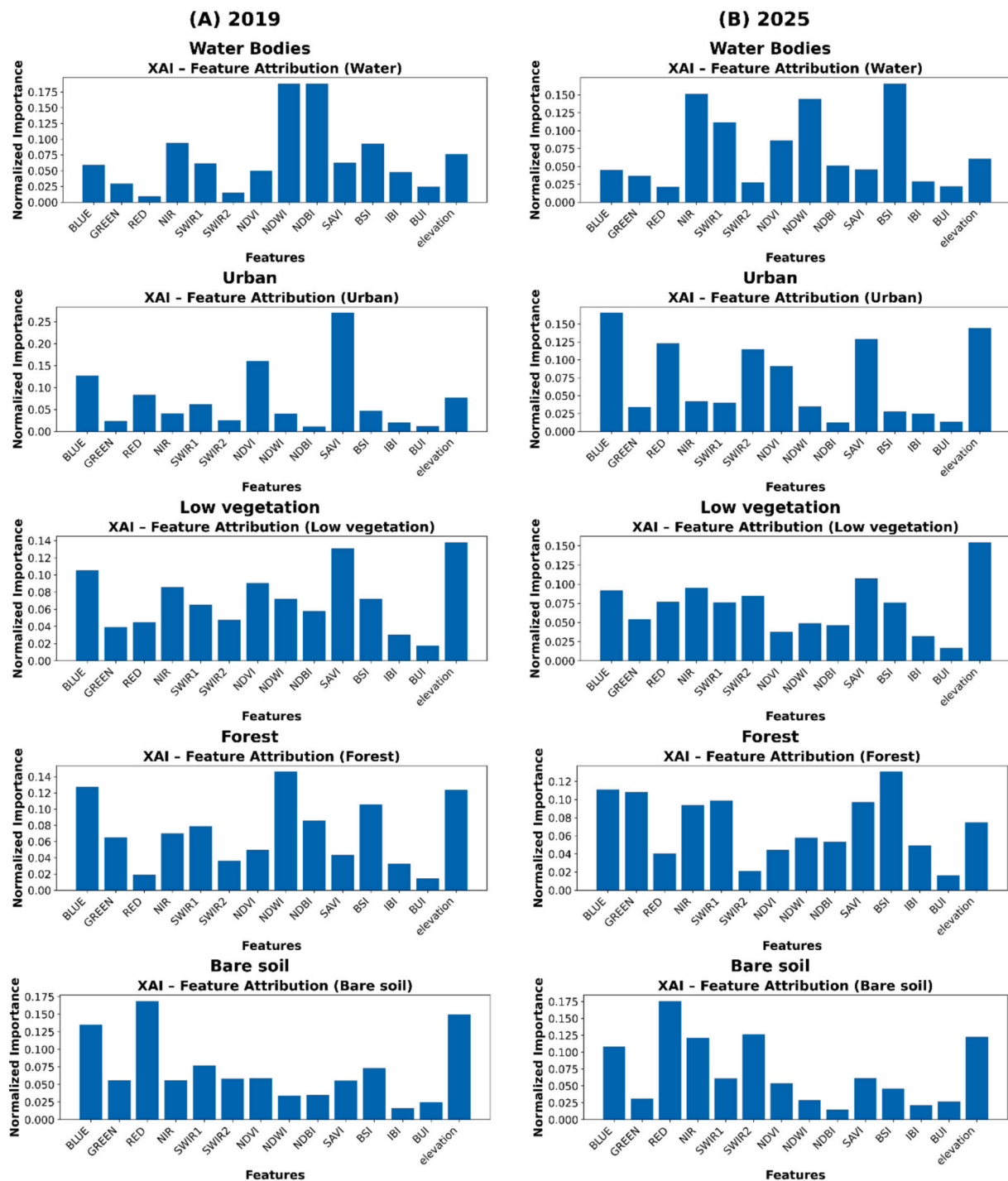


Fig. 12. Class-wise feature attribution (Gradient $\times$ Input) for FS-ViT showing normalized importance of spectral bands and indices for LULC classification in (A) 2019 and (B) 2025. Each row represents a land cover class. The comparison highlights dominant features and their temporal consistency across years.

improve scalability and robustness.

7. Conclusion

This study introduces a Feature-Aware Spectral Vision Transformer (FS-ViT) to classify multi-temporal land use and land cover (LULC) with accuracy and temporal consistency on the mountainous terrain using Sentinel-2 imagery in the spectrally complex area. By treating multi-spectral bands, indices, and elevation as individual tokens and integrating feature-aware reweighting with coarse-to-fine supervision, FS-ViT achieved superior spectral discrimination and temporal stability

compared to conventional classifiers. The proposed model performed better than FA-CNN, FA-KAN, and FA-ExViT in all years considered, with overall accuracies ranging between 96.46% and 98.07% and Kappa coefficients of above 0.93, which was supported by cross-validation. The multi-temporal change analysis indicated that there was an overall forest loss of 6.31 km², with most of the locations changing to low vegetation cover, with a little evidence of vegetation recovery and minimal urban development, which showed a gradual transformation of the landscape by both degradation and regrowth processes. On the whole, the findings prove that feature-conscious spectral transformer learning is a strong and understandable model of long-term LULC

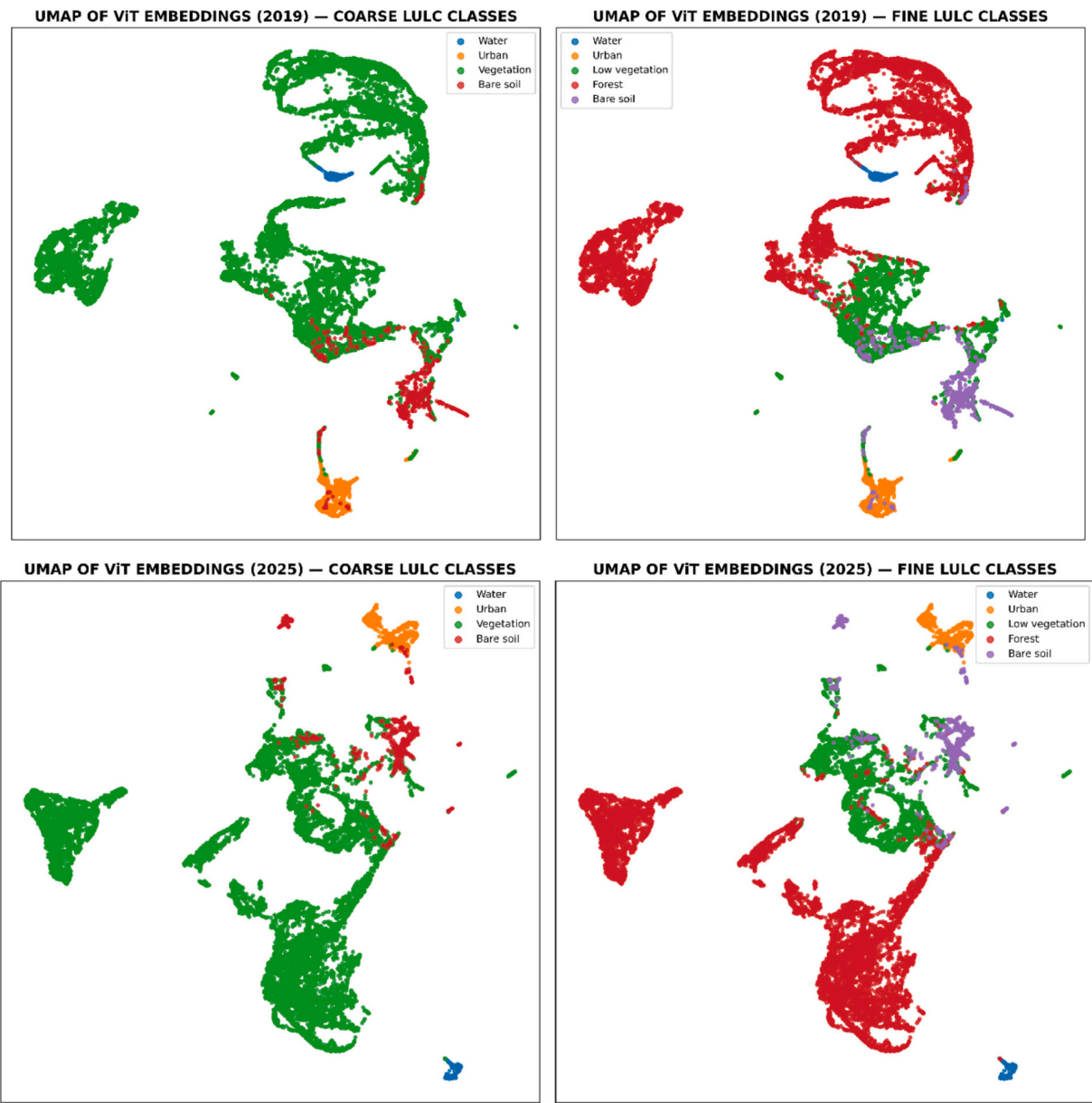


Fig. 13. a. UMAP visualization of latent feature embeddings learned by the FS-ViT model for (a) coarse-level and (b) fine-level LULC classes, illustrating clear class separability and meaningful feature organization of 2019. b. UMAP visualization of latent feature embeddings learned by the FS-ViT model for (a) coarse-level and (b) fine-level LULC classes, illustrating clear class separability and meaningful feature organization of 2025.

mapping and change detection. It is a reliable solution that can be proposed in areas with ecological risks and spectral heterogeneity for environmental monitoring and land-use management.

CRediT authorship contribution statement

C. Saranya Devi: Writing – original draft, Visualization, Software, Resources, Methodology, Formal analysis. **Prabukumar Manoharan:** Writing – review & editing, Validation, Supervision, Investigation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data supporting this study are available from the corresponding author upon reasonable request.

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